

An Entangled Affine Arboreal Image for $z \mapsto z^2$ at Basepoint 2

HCS-C374 / HEN-O358

4 September 2026

Abstract

For the n th preimages of 2 under $z \mapsto z^2$, we determine the splitting field and Galois action uniformly for every $n \geq 3$. The pure radical and cyclotomic fields meet in exactly $\mathbb{Q}(\sqrt{2})$, and this shared quadratic field cuts the full affine group by the character equation $(-1)^b = (2/a)$. The resulting image has index two in $\text{AGL}_1(\mathbb{Z}/2^n\mathbb{Z})$, and adjacent-level restriction is surjective with kernel four.

Keywords: Kummer theory; arboreal Galois representations; cyclotomic fields; affine groups; iterated preimages

中文摘要

本文考察平方映射在基点二之上的逐层原像域。第 n 层的根可用 2^n 次单位根编号；纯根式域与分圆域恰好共享 $\mathbb{Q}(\sqrt{2})$ ，这一个二次交叠迫使仿射参数满足模八的二次特征等式。由此得到指数二的精确 Galois 像，并证明相邻层限制满射且核的阶为四。
关键词：库默理论；迭代原像树；分圆域；仿射群；Galois 作用

1 Preimage fields and the shared quadratic field

Let $m = 2^n$, choose compatible elements

$$\alpha_n = 2^{1/m}, \quad \zeta_m = e^{2\pi i/m},$$

and put

$$R_n = \{\zeta_m^j \alpha_n : j \in \mathbb{Z}/m\mathbb{Z}\}, \quad K_n = \mathbb{Q}(R_n) = \mathbb{Q}(\alpha_n, \zeta_m).$$

These are the level- n vertices and splitting fields of the preimage tree of 2 for $z \mapsto z^2$; see Jones [1] for the general arboreal framework. The present monomial tower is solvable, but it is not the generic full-affine Kummer tower: $\alpha_n^{m/2} = \sqrt{2}$ already lies in the cyclotomic field.

Lemma 1 (Cyclotomic nonsquare). *For every $n \geq 3$, $\sqrt{2}$ lies in $C_n := \mathbb{Q}(\zeta_m)$ but is not a square in C_n .*

Proof. Using the standard cyclotomic description [3], because $\zeta_8 = \zeta_m^{2^{n-3}}$, $\sqrt{2} = \zeta_8 + \zeta_8^{-1} \in C_n$. If $y^2 = \sqrt{2}$ in C_n , then $y = \pm 2^{1/4}$, so $\mathbb{Q}(2^{1/4})$ would be an intermediate field of the abelian Galois extension $C_n/\mathbb{Q}$. Every such intermediate extension is Galois. This is impossible: $X^4 - 2$ is Eisenstein at 2, while the real field $\mathbb{Q}(2^{1/4})$ does not contain its conjugate $i2^{1/4}$. $\square$

Theorem 1 (Radical–cyclotomic intersection). *For all $n \geq 3$,*

$$\mathbb{Q}(\alpha_n) \cap \mathbb{Q}(\zeta_m) = \mathbb{Q}(\sqrt{2}), \quad [K_n : \mathbb{Q}] = 2^{2n-2}.$$

Proof. Over C_n , α_n satisfies $X^{m/2} - \sqrt{2}$. Capelli's binomial criterion [2] reduces irreducibility to excluding a square and, when $4 \mid m/2$, the exceptional shape $\sqrt{2} = -4c^4$. The first is Lemma 1. The second would also make $\sqrt{2}$ a square, namely $(2ic^2)^2$, because $i \in C_n$. Hence $[K_n : C_n] = m/2$, and

$$[K_n : \mathbb{Q}] = (m/2)\varphi(m) = 2^{2n-2}.$$

Eisenstein gives $[\mathbb{Q}(\alpha_n) : \mathbb{Q}] = m$. The compositum formula now gives intersection degree two; the contained field $\mathbb{Q}(\sqrt{2})$ is therefore the whole intersection. $\square$

2 Exact affine image and compatible restrictions

For odd a , let

$$\chi_2(a) = \left(\frac{2}{a}\right) = (-1)^{(a^2-1)/8},$$

viewed modulo 8. Define

$$H_n = \{(a, b) \in (\mathbb{Z}/m\mathbb{Z})^\times \ltimes \mathbb{Z}/m\mathbb{Z} : (-1)^b = \chi_2(a)\},$$

where (a, b) sends j to $aj + b$.

Theorem 2 (Exact image and restriction). *The action on R_n identifies $\text{Gal}(K_n/\mathbb{Q})$ with H_n . This group has order 2^{2n-2} and index two in $\text{AGL}_1(\mathbb{Z}/m\mathbb{Z})$. Under compatible choices $\alpha_n = \alpha_{n+1}^2$ and $\zeta_{2^n} = \zeta_{2^{n+1}}^2$, restriction is*

$$H_{n+1} \longrightarrow H_n, \quad (a, b) \longmapsto (a, b) \pmod{2^n}.$$

It is surjective and its kernel has order four.

Proof. Every automorphism has unique parameters $\sigma(\zeta_m) = \zeta_m^a$ and $\sigma(\alpha_n) = \zeta_m^b \alpha_n$, hence acts by $j \mapsto aj + b$. The two expressions for the shared square root give

$$\sigma(\sqrt{2}) = (-1)^b \sqrt{2} = (\zeta_8^a + \zeta_8^{-a}) = \chi_2(a) \sqrt{2}.$$

Thus the image lies in H_n . There are $\varphi(m)$ choices of a and $m/2$ choices of the forced parity of b , so $|H_n| = 2^{2n-2} = [K_n : \mathbb{Q}]$. The faithful inclusion is an equality.

Squaring the radical and root-of-unity formulas proves coordinate reduction. Each coordinate has two lifts. Reduction preserves a modulo 8 and the parity of b , so all four lifts satisfy the same character equation. This proves surjectivity and the kernel count. $\square$

Round 0 advance: field and compatible-image closure. The intersection, every finite image, and every adjacent restriction map are proved for all levels $n \geq 3$; no finite-level extrapolation occurs.

3 Conclusion

The basepoint 2 produces one persistent quadratic entanglement and no further loss: it cuts the affine image by exactly one character at every level. The resulting compatible group law turns a radical-field question into a complete fixed-root and prime-density theorem.

References

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